North American Academic Research

The TWASP Press

Research

Research on Stock Prediction Based on BiLSTM with Dual Attention Mechanism

Gao Zeyu[1*], Wang Jun[2]

[1] School of Information Engineering, Huzhou University, No. 1 Xue Shi Road, Huzhou, 313000, China
[2] School of Information Engineering, Zhejiang University of Water Resources and Electric Power, No. 508 2nd Avenue, Hangzhou, 310018, China

Article history:
Received: 18 February 2025
Accepted: 19 March 2025
Published Online: 20 March 2025

***Correspondence:**
School of Information Engineering, Huzhou University, No.1 Xue Shi Road, Huzhou, 313000, China

How to cite this article:
Gao Zeyu, Wang Jun (2025). Research on Stock Prediction Based on BiLSTM with Dual Attention Mechanism. North American Academic Research, 8(3),11-19.**doi**:https://doi.org/10.5281/zenodo.15127842

Abstract

Stock price prediction is essential for financial decision-making, yet its complexity poses significant challenges. Traditional statistical models like ARIMA and GARCH rely on strict assumptions and struggle with the high volatility, non-linearity, and noise inherent in stock markets. In contrast, deep learning models, particularly recurrent neural networks (RNNs), have shown superior performance in time series forecasting. However, standard RNNs face limitations in capturing long-range dependencies. To address these challenges, this study proposes a stock prediction model integrating bidirectional long short-term memory (BiLSTM) with a dual attention mechanism. BiLSTM enables the extraction of both past and future dependencies, while the dual attention mechanism separately emphasizes temporal and feature importance, enhancing interpretability. The model is trained and evaluated on real-world stock data, comparing its performance against traditional statistical models and deep learning baselines. Experimental results indicate that the proposed model achieves higher predictive accuracy and better feature selection than conventional approaches. The dual attention mechanism effectively enhances temporal representation and identifies key factors influencing price movements. These findings highlight the advantages of combining sequential modeling with attention mechanisms in financial forecasting. Future work may extend this approach by integrating reinforcement learning for trading optimization and exploring its applicability to multi-asset markets.

Keywords: Deep learning; BiLSTM; Self-Attention mechanism; Stock price prediction

Introduction

Stock price prediction is a crucial yet complex task due to market volatility, non-linearity, and noise. Accurate forecasts help investors make informed decisions and optimize trading strategies. Traditional statistical models, such as ARIMA and GARCH, rely on strict assumptions and often fail to capture complex dependencies in stock prices[1].

With advances in machine learning, deep learning models, particularly RNNs and LSTMs, have shown improved performance in financial time series forecasting[2]. However, standard RNNs struggle with long-term dependencies, limiting their effectiveness in capturing stock price patterns.

Attention mechanisms enhance deep learning models by identifying key time steps and features, reducing irrelevant information. Transformer-based and attention-enhanced LSTMs have improved predictive accuracy[3], yet further refinement is needed to optimize trading decisions.

This study proposes a BiLSTM-based stock prediction model with a dual attention mechanism. BiLSTM captures bidirectional dependencies in stock price movements, while dual attention enhances feature selection and temporal weighting. These improvements aim to enhance forecasting accuracy and trading strategy robustness.

Related Works

In recent years, deep learning has been increasingly applied in the financial sector. By constructing multi hidden layer models and training them on large datasets, deep learning enables complex function approximation and improves prediction accuracy. Long Short-Term Memory (LSTM) networks[4] are among the most popular deep learning models today. With its gating mechanism, LSTM possesses long-term memory capabilities, effectively utilizing long-range sequential information and providing a more robust and reliable solution for financial market prediction[5].

In 2018, Fischer and Krauss[6] applied the LSTM model to predict the fluctuations of the S&P 500 index. Compared to Random Forest (RAF), Deep Neural Networks (DNN), and Logistic Regression (LOG) classifiers, the LSTM model demonstrated superior predictive performance. Di Persio and Honchar (2017) [7] used three different recurrent neural networks to conduct long-term predictions on Google's stock prices. Their results showed that among LSTM, Gated Recurrent Unit (GRU), and basic Recurrent Neural Networks (RNN), the LSTM model performed significantly better in predicting stock prices beyond five trading days, achieving a prediction accuracy of 72%.

However, LSTM can only capture data information in a single direction, which limits its data extraction capability. Its variant, the Bidirectional Long Short-Term Memory (BiLSTM) network, can learn both forward and backward sequences, thereby obtaining more robust temporal features. SIAMI et al.[8] compared the prediction performance of three models—ARIMA, LSTM, and BiLSTM—on financial time series data and concluded that BiLSTM achieved the best predictive results. Yla B et al. (2021)[9] further enhanced BiLSTM by integrating a data decomposition method, constructing the VMD-BiLSTM model. They trained this model on crude oil futures data and ultimately applied it to crude oil price forecasting.

The attention mechanism has achieved great success in recent years. This mechanism assigns corresponding attention weights to different input information, effectively handling dependencies between features and achieving promising results in financial time series forecasting. Zhao and Xue (2021)[10] introduced attention mechanisms into LSTM and CNN models, constructing an LSTM-CNN-CBAM hybrid model, and demonstrated its effectiveness in predicting the SSE Composite Index. Gu Liqiong et al. (2020)[11] applied the attention mechanism to the GRU model and conducted an empirical study on iFLYTEK's stock data, achieving superior experimental results. Lin Jie and Kang Huilin (2020)[12] incorporated the attention mechanism into the LSTM model for stock market prediction in China. Using data from the SSE 50 index, their experiments showed that the proposed model outperformed the standalone LSTM in predictive performance.

Scope of the Study

The aforementioned studies provide the foundation for the selection of research content and methodology in this paper. Considering the non-stationarity, high noise, and volatility clustering characteristics of financial time series, this paper proposes a BiLSTM stock price prediction model, DA-BiLSTM (Dual Attention BiLSTM), which integrates attention mechanisms in both the temporal and feature dimensions.In the temporal dimension, a sliding-window-based time attention mechanism is introduced to dynamically capture the contribution of different historical periods, addressing the traditional BiLSTM's limitation in identifying critical local patterns. In the feature dimension, a hierarchical feature attention network is designed to adaptively allocate feature weights based on feature interaction effects, effectively improving the fusion efficiency of multi-source heterogeneous data. Through the collaborative optimization of these two attention mechanisms, the proposed model not only retains the bidirectional sequential modeling advantage of BiLSTM but also achieves a dual focus on key time points and core feature dimensions.

Model Building

LSTM

The first layer of DA-BiLSTM is the BiLSTM layer. LSTM is a specialized sequential network that controls information retention and transmission through three gating units: the forget gate, the input gate, and the output gate. The forget gate determines which information from the previous cell state C_{t-1} should be discarded. The input gate consists of two neural network layers: a sigmoid layer and a tanh layer. The sigmoid layer takes X_t and H_{t-1} as inputs and outputs I_t, which determines which information needs to be updated. The tanh layer integrates X_t and H_{t-1} to create a new candidate cell state $\tilde{C}_t$ within the range of -1 to 1, which is then used to update the cell state to C_t. The output gate controls the content of C_t that is passed to the current output H_t of the LSTM network.

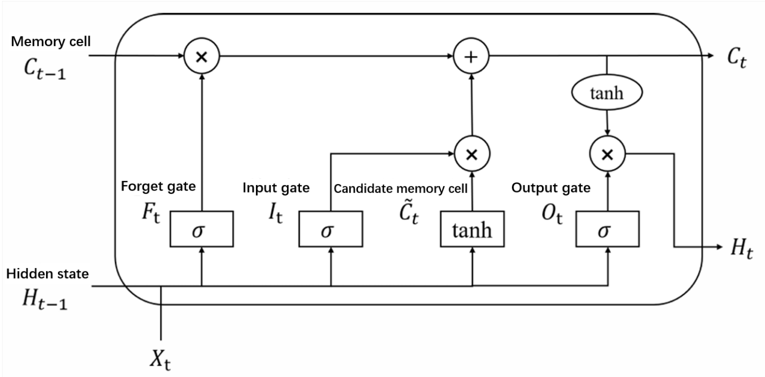

Fig. 1: The Network Structure of LSTM

BiLSTM consists of a forward LSTM and a backward LSTM. The forward LSTM extracts forward features from the input feature sequence, while the backward LSTM captures reverse features from the sequence. The computations of the forward and backward LSTM are shown in Table 1, where W_f, W_i, W_c, and W_o represent the weight matrices of the forget gate, input gate, memory cell state, and output gate, respectively. Similarly, b_f, b_i, b_c, and b_o denote their corresponding biases.Thus, the output y_t at time step t can be computed as follows:

$$y_t = W_y H_t + W_y^* H_t' + b_y \qquad (1)$$

W_y and W_y^* represent the weight matrices for the forward and backward inputs to the hidden layer, respectively, and b_y is the corresponding bias. Since BiLSTM utilizes information from both past and future time steps, it can achieve better performance than LSTM in long-range time series prediction.

Table 1: Computation of Forward LSTM and Backward LSTM

Forward LSTM	Backward LSTM
$F_t = \sigma(W_f[H_{t-1}, X_t] + b_f)$	$F_t = \sigma(W_f[H_{t+1}, X_t] + b_f)$
$I_t = \sigma(W_i[H_{t-1}, X_t] + b_i)$	$I_t = \sigma(W_i[H_{t+1}, X_t] + b_i)$
$\tilde{C}_t = \tanh(W_c[H_{t-1}, X_t] + b_c)$	$\tilde{C}_t = \tanh(W_c[H_{t+1}, X_t] + b_c)$
$C_t = F_t * C_{t-1} + I_t * \tilde{C}_t$	$C_t = F_t * C_{t+1} + I_t * \tilde{C}_t$
$O_t = \sigma(W_o[H_{t-1}, X_t] + b_o)$	$O_t = \sigma(W_o[H_{t+1}, X_t] + b_o)$
$H_t = O_t * \tanh(C_t)$	$H_t' = O_t * \tanh(C_t)$

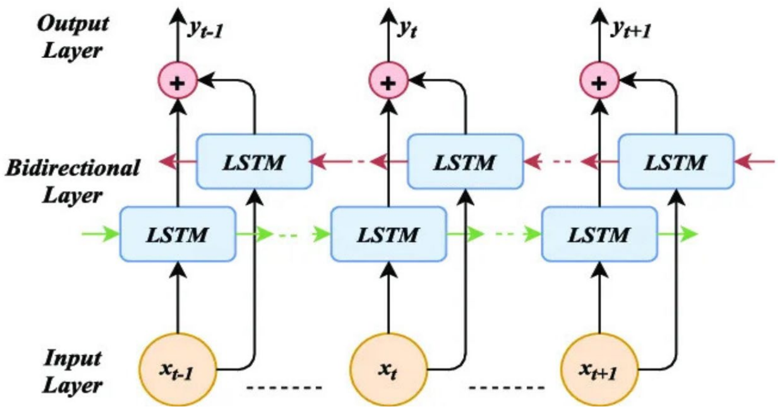

Fig. 2: Schematic Diagram of the BiLSTM Network Structure

Temporal Attention Mechanism

The second layer of DA-BiLSTM is the temporal attention layer. This layer is built on a self-attention mechanism, focusing on identifying key time points that contribute to prediction accuracy in sequential data. It takes the output of the BiLSTM layer as its input. At the attention computation level, a query-key-value matching mechanism is used to dynamically assign temporal weights. First, the hidden states are linearly projected to generate the query, key, and value vectors. Then, the dot product similarity between the query vectors at each time step and all key vectors is computed. After applying Softmax normalization, an attention weight matrix is obtained. The implementation steps are as follows:

The input to the temporal attention layer is $H = [h_1, h_2, \ldots, h_T]$. First, each time step's hidden state h_t undergoes a linear transformation to generate the query vector q_t^{time}, key vector k_t^{time}, and value vector v_t^{time}. The calculation formulas are as follows:

$$q_t^{time} = h_t W_Q^{time} \quad (2)$$

$$k_t^{time} = h_t W_K^{time} \quad (3)$$

$$v_t^{time} = h_t W_V^{time} \quad (4)$$

W_Q^{time}, W_K^{time}, and W_V^{time} are learnable parameter matrices.
Next, for each target time step t, the attention weight with all time steps k is computed using the following formula:

$$\alpha_{t,k}^{time} = \frac{\exp\left(\frac{q_t^{time} \cdot k_k^{time}}{\sqrt{d}}\right)}{\sum_{j=1}^{T} \exp\left(\frac{q_t^{time} \cdot k_j^{time}}{\sqrt{d}}\right)} \quad (5)$$

Finally, the output s_t of the temporal attention layer at time step t is obtained through a weighted sum, calculated as follows:

$$s_t = \sum_{k=1}^{T} \alpha_{t,k}^{time} v_k^{time} \quad (6)$$

The final output of the temporal attention layer can be represented as S, where $S = [s_1, s_2, \dots, s_T]$.
The detailed working mechanism of the temporal attention is shown in Figure 3.

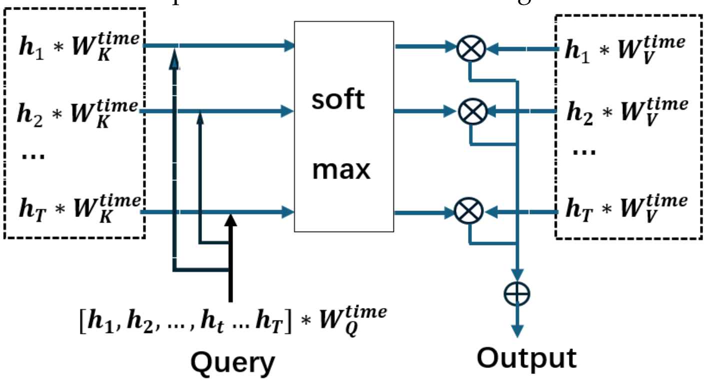

Fig. 3: Illustration of the Temporal Attention Mechanism

Feature-level Attention

The feature attention layer is also based on the self-attention mechanism and takes the output of the temporal attention layer as its input. It focuses on capturing the correlations among multiple features within a single time step. This approach allows for feature extraction in a refined space after temporal filtering, avoiding interference from redundant time steps and accurately capturing the nonlinear relationships between features at the same moment, thus forming a fine-grained representation with both temporal and feature weighting. Its attention computation method is fundamentally similar to that of the temporal attention layer, with the implementation steps as follows:
The feature attention layer takes the output S from the temporal attention layer as its input, where the feature vector at each time step t is represented as $s_t = [s_t^1, s_t^2, \dots, s_t^d]$, with d being the feature dimension. First, each s_t undergoes a linear transformation to generate the query vector q_t^{feat}, key vector k_t^{feat}, and value vector v_t^{feat}. The calculation formulas are as follows:

$$Q_t^{feat} = s_t W_Q^{feat} \quad (7)$$

$$K_t^{feat} = s_t W_K^{feat} \quad (8)$$

$$V_t^{feat} = s_t W_V^{feat} \quad (9)$$

W_Q^{feat}, W_K^{feat}, and W_V^{feat} are learnable parameter matrices.
Next, the attention weights across the feature dimension are computed.

$$\beta_t = softmax\left(\frac{Q_t^{feat} (K_t^{feat})^T}{\sqrt{d}}\right) \quad (10)$$

The weight β_t quantifies the correlation between features.
Finally, a weighted sum is performed to generate the feature-enhanced representation.

$$o_t = \beta_t V_t^{feat} \quad (11)$$

The final output of the feature attention layer can be represented as $O = o_1, o_2, \dots, o_T$.

DA-BiLSTM Network

In addition to the previously mentioned BiLSTM layer, temporal attention layer, and feature attention layer, the DA-BiLSTM network also includes an input layer and a final output layer. The specific structure is as follows:

Input Layer: Receives time-series data with a sequence length of N and k features per step. The input dimension is $[B, N, k]$, where B is the batch size.

BiLSTM Layer: A bidirectional LSTM structure with concatenated outputs from both directions. Each direction has h_{size} units. The layer returns the hidden state outputs for all time steps, with an output dimension of $[B, N, h_{size} * 2]$.

Dropout Layer: Applied to the hidden state outputs of the BiLSTM layer with a dropout probability of p to prevent overfitting.

Temporal Attention Layer: Takes the hidden states from the BiLSTM layer as input with a dimension of $[B, N, h_{size} * 2]$. It models temporal dependencies through the self-attention mechanism, maintaining the same output dimension of $[B, N, h_{size} * 2]$.

Feature Space Attention Layer: Takes the output of the temporal self-attention layer as input. It dynamically computes feature weights to enhance key feature dimensions. The output dimension remains the same as the input: $[B, N, h_{size} * 2]$.

Fully Connected Output Layer: Uses the tanh activation function with a single unit to directly output the predicted closing price for the next day. The output dimension is $[B, 1]$.

The schematic diagram of the DA-BiLSTM network is as follows.

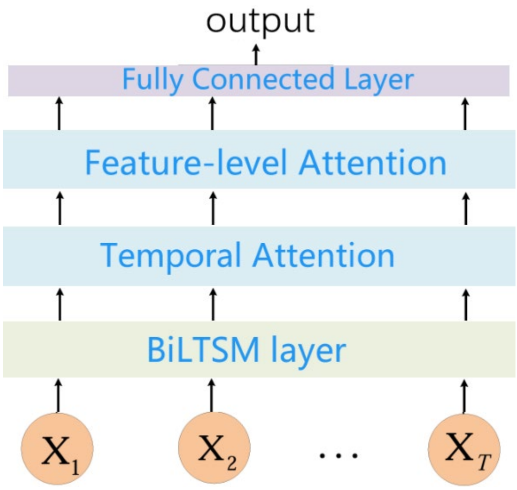

Fig. 4: Schematic Diagram of the DA-BiLSTM Network

Empirical Analysis

Data Sources and Processing

This study selects Apple's stock market data for empirical analysis. As a leading company in the industry, its stock price index is highly reliable and representative, making it a strong reflection of overall market conditions and providing a solid data foundation for research.

The stock data is sourced from Yahoo Finance, covering the period from January 1, 2015 to December 31, 2022 with a total of 1,946 days of data. To construct the stock dataset, three categories of indicators are selected: volume-price indicators, technical indicators, and macroeconomic indicators, detailed as follows:

Volume-Price Indicators: Opening price, closing price, highest price, lowest price, trading volume, and turnover rate.

Technical Indicators: 5-day moving average, Relative Strength Index, and Bollinger Bands. The Bollinger Bands consist of the middle band, upper band, and lower band.

Macroeconomic Indicators: The daily closing prices of the Dow Jones Industrial Average, Standard & Poor's 500 Index, and NASDAQ Composite Index are used as macroeconomic indicators.

For model training and evaluation, 80% of the data is used as the training set, while the remaining 20% is used as the test set. Before fitting the model to the data, normalization is applied to accelerate gradient descent and improve computational accuracy. The data is linearly scaled to the range of [0,1].

Evaluation Metrics

To ensure a comprehensive evaluation of the models, this study employs two error metrics: Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). These metrics are used to assess the predictive performance of the proposed model compared to baseline models. MAE measures the absolute magnitude of errors, while MAPE evaluates the relative error. A lower value for both metrics indicates higher prediction accuracy.

Parameter Settings

The initial parameter settings for the model are summarized in the following table:

Table 2: Initial Parameter Settings

Parameter	Parameter Value

Window size	15
Epoch,Batch size	100,64
BiLSTM layer units	64
Fully connected layer units	64
Initial learning rate	0.002
Optimizer	Adam
Dropout rate	0.2

Based on existing research experiments[13], it has been found that window size, BiLSTM hidden layer units, and fully connected layer units significantly impact the predictive performance of the model. Therefore, a parameter tuning experiment is conducted to determine the optimal values, using Mean Squared Error (MSE) as the loss function.

(a) BiLSTM layer units and Fully Connected Layer Units

BiLSTM layer units and fully connected layer units are not independent; they have a strong interdependent relationship. The number of BiLSTM units determines the model's ability to capture contextual information in sequence data. A larger number of units allows richer sequence feature extraction but may introduce excessive noise and redundant information. The fully connected layer refines and maps the features extracted by BiLSTM and the dual-attention layers, transforming them into the final prediction. To find the best balance between these two parameters and optimize model performance, a grid search is used while keeping other parameters constant. Other parameters remain fixed during the parameter tuning experiment, with MAE used as the evaluation metric.

Table 3: MAE Values Under Different Parameter Combinations

(BiLSTM layer units，FC layer units)	MAE
（64，64）	72.511
（128，64）	66.847
（256，64）	70.903
（128，128）	59.279
（256，128）	51.371
（256，256）	53.365

The experiment results show that the optimal parameter combination is (256, 128), where MAE reaches its minimum value, indicating the best predictive performance. Therefore, the BiLSTM layer units are set to 256, and the fully connected layer units are set to 128.

(b) Window Size Selection

The size of the sliding window represents the length of input data during training, which in this study refers to the amount of historical data used to predict future closing prices. The window size significantly affects prediction performance. A window that is too small may result in insufficient sequential information capture, while an excessively large window may introduce irrelevant information that interferes with extracting useful patterns, ultimately affecting the quality of high-level feature extraction. Therefore, this study determines the optimal window size through experimental evaluation of the model's performance under different window sizes. The experiment sets the window sizes to 5, 10, 15, and 20, examining the model's ability to predict the closing prices for the next three days. Other parameters remain fixed during the parameter tuning experiment, with MAE used as the evaluation metric.

Table 4: MAE Values for Different Window Sizes

Window Size	5	10	15	20
MAE	56.341	43.178	47.523	48.896

The comparative experiment shows that the evaluation metric value first decreases and then increases as the window size grows. The MAE reaches its minimum when the window size is 10, indicating the best predictive performance of the model at this setting. Thus, the optimal window size is set to 10.

Based on these experimental results, the final parameter settings of the DA-BiLSTM prediction model are summarized in the following table:

Table 5: Table: Final Parameter Settings

Parameter	Parameter Value
Window size	10
Epoch,Batch size	100,64
BiLSTM layer units	256
Fully connected layer units	128
Initial learning rate	0.002
Optimizer	Adam

Dropout rate	0.2

Comparative Experiment Results Analysis

To verify the superiority of the proposed DA-BiLSTM hybrid model in stock prediction, the standard BiLSTM model, the BiLSTM model incorporating only time attention (TA-BiLSTM) and the BiLSTM model incorporating only feature attention (FA-BiLSTM) were established as baseline models for comparison under the same conditions.

The stock price prediction results are shown in the following figures.

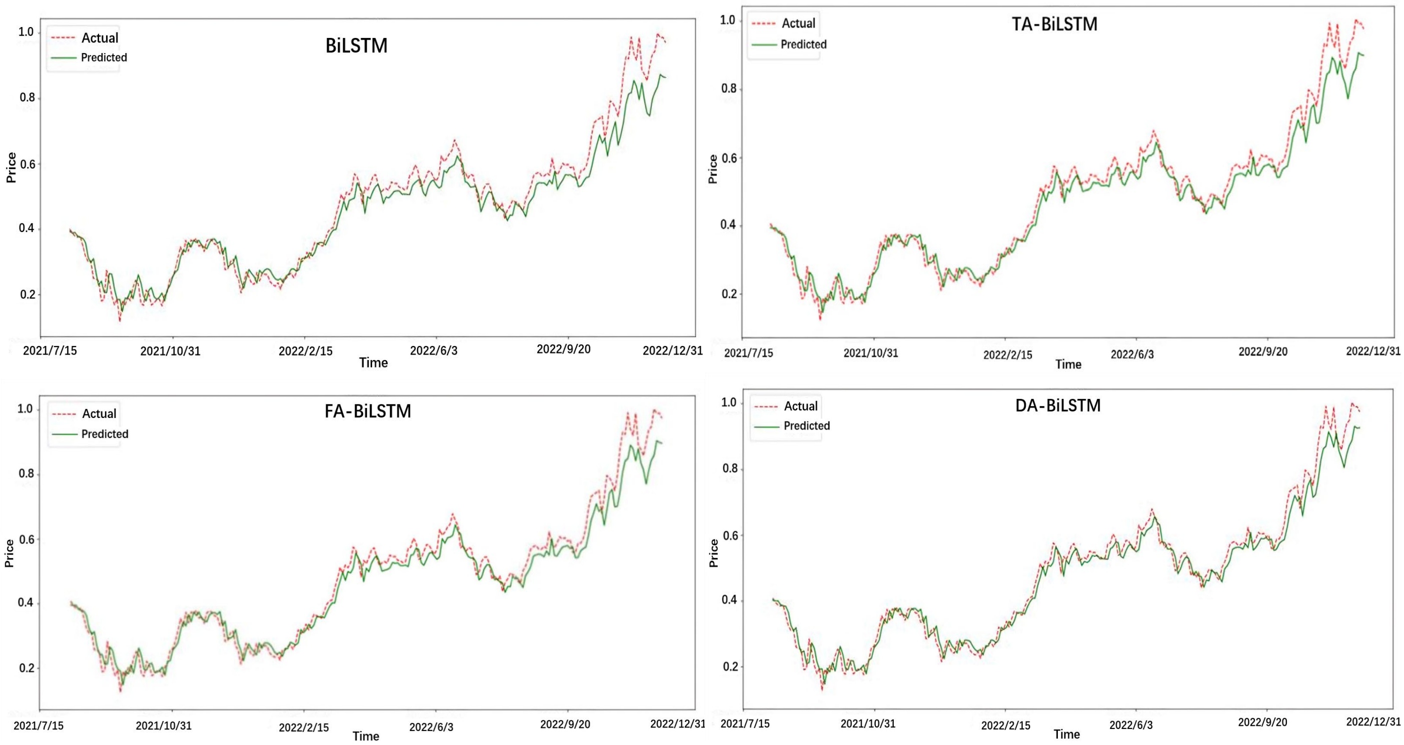

Fig. 5: Prediction Results of Different Models

In the figures, the X-axis represents the date, while the Y-axis represents the normalized closing price. The red dashed line indicates the actual closing price curve, whereas the green solid line represents the predicted closing price curve.

From the prediction curves, it is evident that DA-BiLSTM achieves a more significant improvement compared to both TA-BiLSTM and FA-BiLSTM. Although TA-BiLSTM and FA-BiLSTM both enhance the performance of BiLSTM, TA-BiLSTM performs slightly better. This could be because BiLSTM, when handling time-series data, inherently captures some feature correlations, whereas FA-BiLSTM fails to reallocate the weights of BiLSTM's hidden layer outputs according to the importance of different time steps.

The evaluation metrics of the four models' prediction results are presented in Table 6.

Table 6 : Model Performance Comparison

Model	MAE	MAPE
BiLSTM	0.397	0.0585
TA-BiLSTM	0.328	0.0526
FA-BiLSTM	0.352	0.0543
DA-BiLSTM	**0.281**	**0.0469**

The data in the table clearly demonstrate that DA-BiLSTM exhibits a significant advantage over BiLSTM, TA-BiLSTM, and FA-BiLSTM. Regarding the MAE (Mean Absolute Error) metric, DA-BiLSTM achieves a value of 0.291, which is lower than that of the other models. This indicates a smaller average absolute error, meaning that the predicted values deviate less from the actual values on average. For the MAPE (Mean Absolute Percentage Error) metric, DA-BiLSTM attains a value of 0.0473, which is also lower than that of the other models. This suggests that the relative deviation between the predicted and actual values is smaller, further confirming the model's superior predictive accuracy.

Conclusion

To address the nonlinear characteristics of financial time series, this study proposed a BiLSTM model integrated with both time and feature attention mechanisms. Through comparative experiments, the model was applied to predict Apple's stock closing prices, evaluating its effectiveness. The final conclusions drawn from this study are as follows:

Incorporating time and feature attention mechanisms into BiLSTM significantly enhances its ability to predict financial time series.

Author Contributions: At first page.
Approval: All authors have read and agreed to the published version of the manuscript.
Funding: This research received no external funding.
Institutional Review Board Statement: Not applicable.
Informed Consent Statement: Not applicable.
Data Availability Statement: Not applicable
Acknowledgments: Not Mentioned.
Conflicts of Interest: The authors declare no conflict of interest.

References

[1] Petrică, Andreea-Cristina, Stelian Stancu, and Alexandru Tindeche. Limitation of ARIMA models in financial and monetary economics[J]. Theoretical & Applied Economics. 2016, 23(4).
[2] S. Selvin, R. Vinayakumar, E. A. Gopalakrishnan, V. K. Menon and K. P. Soman. Stock price prediction using LSTM, RNN and CNN-sliding window model[C] // International Conference on Advances in Computing, Communications and Informatics (ICACCI). Udupi, India, 2017, pp. 1643-1647
[3] Chen, Shun, and Lei Ge. Exploring the Attention Mechanism in LSTM-Based Hong Kong Stock Price Movement Prediction[J]. Quantitative Finance. 2019, 19(9):1507–15..
[4] Hochreiter S, Schmidhuber J. Long Short-Term Memory[J]. Neural Computation. 1997, 9(8):1735-1780.
[5] YANG Li et al. Research on recurrent neural network [J]. Journal of Computer Applications. 2018, 38(S2).
[6] FISCHER T, KRAUSS C. Deep learning with long short-term memory networks for financial market predictions [J]. European Journal of Operational Research. 2018, 270(2):654-669.
[7] Di Persio L,Honchar O. Recurrent neural networks approach to the financial forecast of Google assets[J]. International Journal of Mathematics and Computers in Simulation. 2017, 11: 7–13.
[8] Siami-Namini, Sima et al. A Comparative Analysis of Forecasting Financial Time Series Using ARIMA, LSTM, and BiLSTM[J]. arXiv preprint arXiv: 1911.09512, 2019.
[9] Yla B, Sj C, Xl A, et al. The role of news sentiment in oil futures returns and volatility forecasting: Data decomposition based deep learning approach[J]. Energy Economics. 2021: 105140.
[10] ZHAO Hongrui, XUE Lei. Research on Stock Forecasting Based on LSTM-CNN-CBAM Model[J]. Computer Engineering and Applications. 2021, 57(03): 203-207.
[11] GU Liqiong et al. An Attention-based GRU Model for Stock Predicting[J]. Systems Engineering. 2020, 38(05): 134-140.
[12] LIN Jie, KANG Huilin. Attention-Based LSTM for Stock Price Movements Prediction [J]. Shanghai Management Science. 2020, 42(01): 109-115.
[13] NELSON D.M.Q. et al. Stockmarket's pricemovement prediction with LSTM neural networks[C] // International Joint Conference on Neural Networks. IEEE, 2017:1419-1426.

Gao Zeyu. Master student of Huzhou University, China.
Prof Wang Jun received PhD from Shanghai University of Finance and Economics, China.